

Economics

Higher level

Paper 2

SME Predicted Paper – Advanced

1 hour 45 minutes

Instructions to candidates

- Do not open this paper until instructed to do so.
- You are permitted access to a calculator for this paper.
- Unless otherwise stated in the question, all numerical answers must be given exactly or correct to two decimal places.
- You must show all your working.
- Answer one question.
- Use fully labelled diagrams and references to the text/data where appropriate.
- The maximum mark for this examination paper is [40 marks].

Answer one question.

1. Read the extracts and answer the questions that follow.

Text A — Nigeria: The Paradox of Plenty

1. Nigeria is Africa's largest economy by population (220 million) and one of its largest by GDP (US\$477 billion in 2022). The country is rich in oil: petroleum accounts for around 6–9% of GDP, 60% of government revenue, and over 85% of export earnings.
2. Despite substantial oil wealth, Nigeria's per-capita income growth has been disappointing. Real GDP per capita actually fell in US dollar terms between 2014 and 2022, as the naira depreciated sharply and population growth (2.4% per year) outpaced real GDP growth. Over 40% of Nigerians live below the national poverty line.
3. Inflation accelerated to 22% in 2023, driven by food prices (up 26%), fuel-subsidy removal, and the collapse of the parallel exchange rate. The Central Bank of Nigeria (CBN) raised its monetary policy rate from 11.5% in May 2022 to 18.75% by July 2023, one of the most aggressive tightening cycles in recent Nigerian history.
4. Nigeria has long suffered from what economists call Dutch disease: oil exports drive up the real exchange rate, making non-oil tradables (manufactures, agriculture) uncompetitive. Manufacturing as a share of GDP has declined from 14% in the 1980s to around 9% today.
5. The recently-unified Nigerian naira depreciated from around 465/US\$ to over 770/US\$ between June and August 2023, after the new administration removed a long-standing multiple-exchange-rate system. The move has been praised by economists but has fuelled short-term inflation.

Text B — Beyond Oil: Fiscal Reform and Subsidy Removal

1. For decades Nigeria subsidised fuel heavily, holding pump prices artificially low. The subsidy cost the government around US\$10 billion a year and encouraged smuggling to neighbouring countries where prices are higher.
2. In 2023 the government fully removed the fuel subsidy. Pump prices roughly tripled, generating significant fiscal savings but also triggering strikes, protests and further inflation.
3. The government argues that fiscal savings from subsidy removal (projected at around 2% of GDP) can be redirected to social safety nets, infrastructure and education. Critics note that Nigeria's weak institutional capacity and history of corruption may undermine the promised reinvestment.
4. Nigeria is also seeking to diversify its economy. The agricultural sector employs 35% of the workforce, and digital services (financial technology, software) have been a bright spot, with FDI flowing into companies such as Flutterwave. However, these sectors remain small relative to oil.
5. Nigeria's debt service now consumes around 96% of federal revenue — a level widely regarded as unsustainable. External debt stood at US\$41 billion in mid-2023.

	2014	2019	2022
GDP (US\$ billion)	568.5	448.1	477.4
GDP per capita (US\$)	3,222	2,229	2,163
Inflation rate (%)	8.1	11.4	18.8

	2014	2019	2022
Policy rate (% end of year)	13.0	13.5	16.5
Exchange rate (NGN / US\$, year-average)	164	306	435
Manufacturing (% of GDP)	9.8	9.1	8.9

Table 1: Selected indicators for Nigeria, 2014–2022

Question 1 continued

(a)(i) Define the term Dutch disease indicated in bold in Text A, Paragraph 4. **[2]**

(a)(ii) Define the term subsidy indicated in bold in Text B, Paragraph 1. **[2]**

(b)(i) Using Table 1, calculate the percentage change in Nigeria's GDP per capita between 2014 and 2022. **[3]**

(b)(ii) Using Table 1, calculate the real policy interest rate in 2022 (to 2 dp) using the Fisher approximation. **[2]**

(c) Using a subsidy diagram, explain the effect of removing the fuel subsidy (Text B, Paragraph 2) on the Nigerian fuel market. **[4]**

(d) Using an AD/AS diagram, explain how the combined effects of subsidy removal and naira depreciation (Text A, Paragraph 5) could affect Nigeria's price level and real output. **[4]**

(e) Using an exchange-rate diagram, explain how Nigeria's heavy dependence on oil exports (Text A, Paragraph 1) can cause appreciation of the naira's real exchange rate — a mechanism central to Dutch disease. **[4]**

(f) Using a Laffer-curve or appropriate diagram, explain the relationship between Nigeria's debt-service burden (Text B, Paragraph 5) and its ability to finance development expenditure. **[4]**

(g) Using information from the text/data and your knowledge of economics, evaluate the view that the 2023 policy package (subsidy removal, FX unification, monetary tightening) will promote sustainable development in Nigeria. **[15]**

2. Read the extracts and answer the questions that follow.

Text A — Germany: Deindustrialisation or Green Renewal?

1. Germany, Europe's largest economy with a GDP of US\$4.07 trillion in 2022, has long been synonymous with advanced manufacturing — particularly automobiles, machinery and chemicals. Manufacturing accounts for around 19% of GDP, well above the OECD average of 14%.
2. The German economic model has relied on three pillars: cheap Russian energy, cheap Chinese demand for German capital goods, and a highly skilled industrial workforce. Two of these pillars have come under severe pressure since 2022.
3. The Russian invasion of Ukraine led to a cut-off of pipeline natural gas imports from Russia. Wholesale gas prices in Europe rose tenfold at their peak in August 2022. Energy-intensive industries — chemicals, steel, glass, ceramics — cut output sharply or relocated production abroad. BASF, the world's largest chemical company, announced permanent capacity closures in Germany.
4. At the same time, China's slowing growth and rise as a direct competitor in key export markets (autos, machinery) has weakened German external demand. Exports to China have fallen in real terms for three consecutive quarters by end-2023.
5. Germany slipped into a mild recession in 2023, with real GDP contracting by 0.3%. Inflation reached a peak of 8.8% in October 2022 before gradually easing.

Text B — The Green Transition

1. The German government has responded with an ambitious industrial policy: the Energiewende (energy transition) and a €200 billion 'economic shield' to subsidise energy costs for firms and households. The 2023 Inflation Reduction Act (IRA) in the United States has also intensified pressure on European policymakers to match US subsidies for green industries.
2. Germany plans to reach net-zero emissions by 2045. Renewables now account for over 50% of electricity generation, up from 17% in 2010. Investment in battery plants, green hydrogen, and wind/solar capacity is accelerating, aided by federal subsidies.
3. Some economists argue that the energy-price shock and structural shifts amount to a 'de-industrialisation' risk: once production capacity leaves Germany, it rarely returns. Others see it as a necessary transformation toward a cleaner, higher-value economy.
4. Labour market conditions remain tight. The unemployment rate is 5.7% but there are around 1.7 million vacant jobs, and the German population is ageing rapidly, with the old-age dependency ratio projected to rise from 34% to 50% by 2040.
5. Germany continues to run a current account surplus, but this has narrowed from a peak of 8.9% of GDP in 2015 to 4.2% in 2022, as energy import costs surged.

	2013	2018	2022
GDP (US\$ trillion)	3.75	3.98	4.07
Real GDP growth (%)	0.4	1.0	1.8
Manufacturing (% of GDP)	22.0	21.4	19.1

	2013	2018	2022
Inflation rate (%)	1.5	1.7	6.9
Current account (% of GDP)	6.7	7.8	4.2
Renewables (% of electricity)	24	35	52

Table 1: Selected indicators for Germany, 2013–2022

Question 2 continued

(a)(i) Define the term industrial policy indicated in bold in Text B, Paragraph 1. **[2]**

(a)(ii) Define the term current account surplus indicated in bold in Text B, Paragraph 5. **[2]**

(b)(i) Using Table 1, calculate the percentage change in Germany's current account balance (% of GDP) between 2013 and 2022. **[3]**

(b)(ii) Using Table 1, calculate the absolute change in the share of renewables in electricity between 2013 and 2022 (percentage-point change). **[2]**

(c) Using an AD/AS diagram, explain how the 2022 gas-price shock (Text A, Paragraph 3) could affect Germany's price level and real GDP. **[4]**

(d) Using a subsidy diagram, explain how the €200 billion 'economic shield' (Text B, Paragraph 1) could affect the energy market and household welfare. **[4]**

(e) Using an appropriate diagram, explain how an ageing population and tight labour market (Text B, Paragraph 4) could affect Germany's long-run potential output. **[4]**

(f) Using an exchange-rate diagram, explain how a narrowing current account surplus (Text B, Paragraph 5) could affect the euro. **[4]**

(g) Using information from the text/data and your knowledge of economics, evaluate the view that Germany's Energiewende is the most effective strategy for restoring long-run economic growth. **[15]**